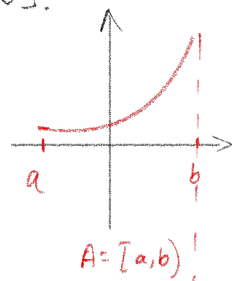
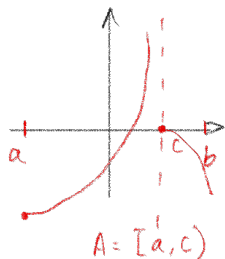
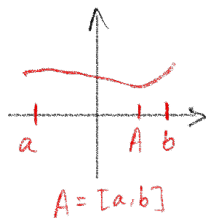


PF of Boundness Thm

If f is cts. on $[a, b]$, then f is bounded above $[a, b]$.

Idea



PF. Let $A := \{a \leq x \leq b : f \text{ is bounded above on } [a, x]\}$.

• $A \neq \emptyset$ b/c $a \in A$

• A is bounded above b/c b is an upper bound of A by construction

$\Rightarrow A$ has a LUB by (P13) and call it α .

Suppose for a contradiction that $\alpha \neq b \Rightarrow \alpha < b$

Let $\varepsilon = 1$. Since f is cts. on $[a, b]$, it is cts. at $\alpha \in [a, b]$

So $\exists \delta > 0$ s.t. whenever $0 < |x - \alpha| < \delta$

$$|f(x) - f(\alpha)| < \varepsilon = 1$$

$$\Leftrightarrow -1 < f(x) - f(\alpha) < 1$$

$$f(\alpha) - 1 < f(x) < f(\alpha) + 1$$

So $f(x)$ is bounded on $(\alpha - \delta, \alpha + \delta)$ and also bounded on $[a, \delta]$.

So it is bounded on $[a, \alpha + \delta]$.

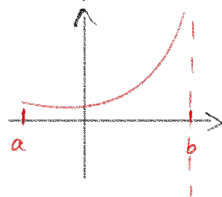
So $\frac{\alpha + \delta}{2} \in A$, α is not an UB, left alone the LUB, contradiction $\Rightarrow \alpha = b$

Now it remains to rule out $A = [a, b)$. We will use continuity of f .

For $\varepsilon = 1$, f is bounded above $B+1$, whenever B is an upper bound for $\{f(x) : x \in [a, b)\}$

$\exists \delta > 0$ s.t. whenever $0 < |x-b| < \delta$, $|f(x) - B| < 1$

So f is also bounded by $B+1$ at $b \Rightarrow b \in A$



Pf of EVT: If f is cts. on $[a, b]$, $\exists c \in [a, b]$ s.t. $f(x) \leq f(c) \forall x \in [a, b]$.

$$\Rightarrow A = \{f(x) : x \in [a, b]\}$$

• $A \neq \emptyset$ (for eg. $f(a), f(b), \dots \in A$).

• A bounded above by boundedness thm

$\Rightarrow A$ has the LUB α .

Suppose for a contradiction that $f(x) \neq \alpha \forall x \in [a, b]$.

Construct a new function $g(x) = \frac{1}{\alpha - f(x)}$, defined on $[a, b]$.

Since $f(x) \neq \alpha$, $g(x)$ is cts. on $[a, b]$.

$\Rightarrow$ By Boundedness Thm, $g(x)$ is also bounded on $[a, b]$.

However, α is the LUB of all $f(x)$ where $x \in [a, b]$

$\Rightarrow$ any $\varepsilon > 0$, $\exists x \in [a, b]$ s.t. $f(x) = \alpha - \varepsilon$.

$\rightarrow$ By IVT + continuity of f .

$$\text{For this } x_\varepsilon, g(x_\varepsilon) = \frac{1}{\alpha - f(x_\varepsilon)}$$

$$= \frac{1}{\alpha - (\alpha - \varepsilon)} = \frac{1}{\varepsilon} \rightarrow \infty \text{ as } \varepsilon \rightarrow 0$$

So it contradicts boundedness thm.

i.e. $\exists c \in [a, b]$ s.t. $f(c) = \alpha$,



Q: In the pf of IVT, We used $A = \{x \in [a, b] : f \leq 0 \forall [a, x]\}$.

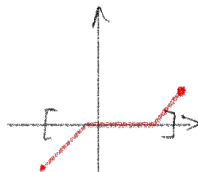
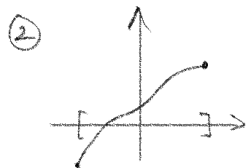
So this gives us the location of the first pt of intersection of f w/ x -axis.

① Can you modify the pf so that you can find the last part of intersection?

② _____ " _____ 2nd part of intersection?

Soln ① Instead of A , use $A' = \{x \in [a, b] : f > 0 \text{ for all values in } [x, b]\}$.

+ use GLB property.



The 2nd pt of intersection might not exist.

Def. $A \subseteq \mathbb{R}$ is dense if for any open $I \subseteq \mathbb{R}$, $I \cap A \neq \emptyset$

thm. $\mathbb{Q}$ is dense in $\mathbb{R}$.

Idea. Let $I = (x, y)$ WTS. $\exists \frac{m}{n} \in (x, y)$

$$x < \frac{m}{n} < y$$

$$\Leftrightarrow nx < m < ny$$

i.e. If dilate (x, y) by a large enough integer, this interval will eventually

contain an integer.

pf. Since $y-x > 0$

$$\exists n \text{ s.t. } y-x > \frac{1}{n} \Rightarrow ny-nx > 1$$

It contains an integer $m \Rightarrow \frac{m}{n} \in (x, y)$.